Retail Business Performance & Profitability Analysis

Dataset: Sample - Superstore (Kaggle) **Records:** 9,994 **Tools:** SQL, Python, Tableau

1. Introduction

Retail businesses generate large amounts of transaction data. Analysing this data helps identify profitable product categories, loss-making sub-categories, regional sales patterns and changes in sales over time. This project analyses the Sample - Superstore dataset to understand business performance and provide data-based insights that can support better business decisions.

2. Abstract

This project focuses on analysing retail transaction data to evaluate sales and profitability. The dataset contains 9,994 records with information such as order dates, regions, product categories, sales, quantity, discounts and profit. Python (Pandas) was used to inspect and clean the data, convert date fields, remove duplicate records and create additional fields such as order year, order month and profit margin. SQL queries were used to calculate total sales and profit, compare profit by category, examine sales by region, identify loss-making sub-categories and study average profit across discount levels. Tableau was used to build an interactive dashboard containing category profit, regional sales, sub-category profit and monthly sales trend charts, with Region, Category and Order Year filters. The analysis highlights areas of strong performance and areas that may require attention.

3. Tools Used

- **Python:** Pandas for data loading, cleaning and analysis; Matplotlib and Seaborn for charts.
- **SQLite / DB Browser for SQLite:** storing the cleaned data and running SQL queries.
- **Tableau:** creating an interactive dashboard with charts and filters.
- **Dataset:** Sample - Superstore CSV downloaded from Kaggle.

4. Steps Involved in Building the Project

1. Downloaded the Sample - Superstore dataset and loaded it into Python.
2. Inspected the dataset, checked missing values, converted Order Date and Ship Date to date format, removed duplicate rows and created Order Year, Order Month and Profit Margin fields.
3. Used Python to calculate overall sales, profit and quantity, and to compare profit by category and sub-category, sales by region and average profit by discount.
4. Imported the cleaned CSV into SQLite and wrote SQL queries for business performance, category profitability, regional sales, loss-making sub-categories and average profit by discount.
5. Connected the cleaned dataset to Tableau and created four visualisations: Profit by Product Category, Sales by Region, Profit by Sub-Category and Monthly Sales Trend.
6. Combined the visualisations into an interactive dashboard and added Region, Category and Order Year filters. Saved the dashboard as a Tableau Packaged Workbook.

5. Key Findings and Conclusion

The dataset records total sales of $2,297,200.86 and total profit of $286,397.02. Technology generated the highest profit among the three product categories, while Furniture generated the lowest. The West region recorded the highest sales. At sub-category level, Tables showed the largest overall loss, while Copiers generated the highest profit. Higher discount levels were often associated with lower average profit in this dataset; this indicates an association and does not by itself prove that discounts caused the losses. The project demonstrates how Python, SQL and Tableau can be combined to turn transaction data into understandable business insights. One limitation is that the dataset does not contain stock-on-hand or inventory-days information. Therefore, actual inventory turnover and stock ageing could not be calculated; quantity sold is used only as a demand-related measure. Adding inventory records would allow a more direct inventory analysis in future.